

Introduction

The Big Bang has left echoes of sound waves from the nascent universe in the form of a characteristic pattern known as **baryon acoustic oscillations (BAOs)**.

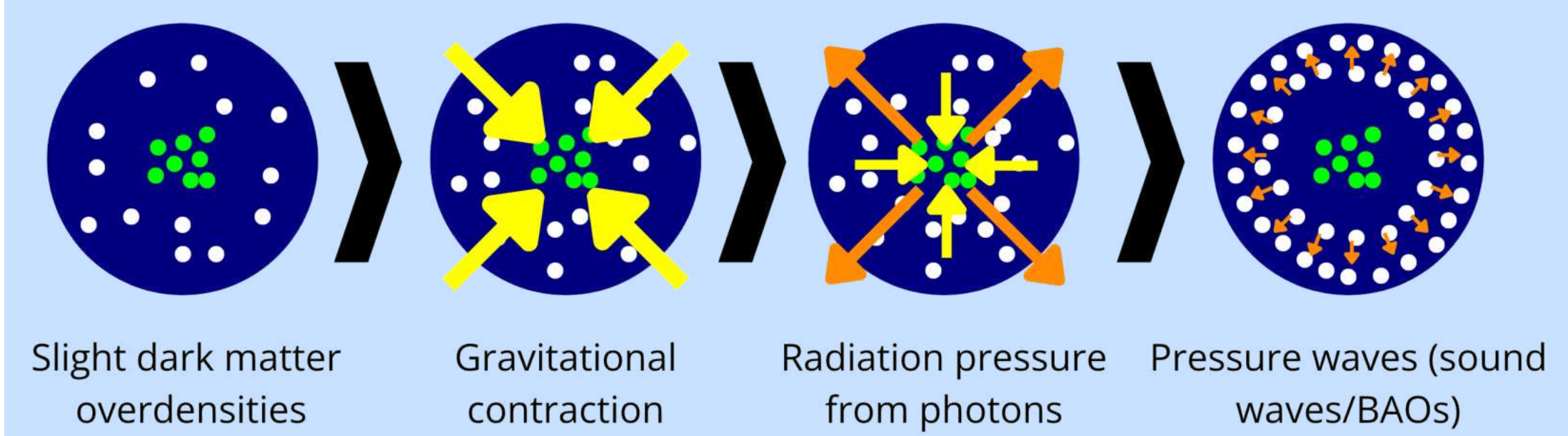


Figure 1. Graphic depicting the formation of BAOs. Graphic created by finalist using Canva, 2025.

Basics of Baryon Acoustic Oscillations

- **Matter-light** interactions (Figure 1)
- Photon decoupling **halts** outward pressure
- BAOs **freeze** after **380,000 years**

BAOs left a distinctive imprint on the **distribution of matter in the universe**, which we can detect in the form of a **characteristic peak in the separations of galaxies**. The location of the peak can be used as a **“standard ruler.”**

The **Hubble constant**, denoted by H_0 , describes the **expansion rate of the universe**, measured in $\text{km s}^{-1} \text{Mpc}^{-1}$. BAOs are linked to the **Hubble constant** as their size at a certain point in time **depends on the expansion rate** of the universe.

The **density parameters** Ω_Λ and Ω_m quantify the ratios of dark energy and dark matter to the critical density that would result in a spatially-flat universe.

Purpose

Past studies such as Eisenstein et al. (2005) have detected the BAO signal at **low redshift** using the Sloan Digital Sky Survey data releases. A new instrument, the **Dark Energy Spectroscopic Instrument (DESI)** has released its first official data release (**DR1**), which awaits analysis.

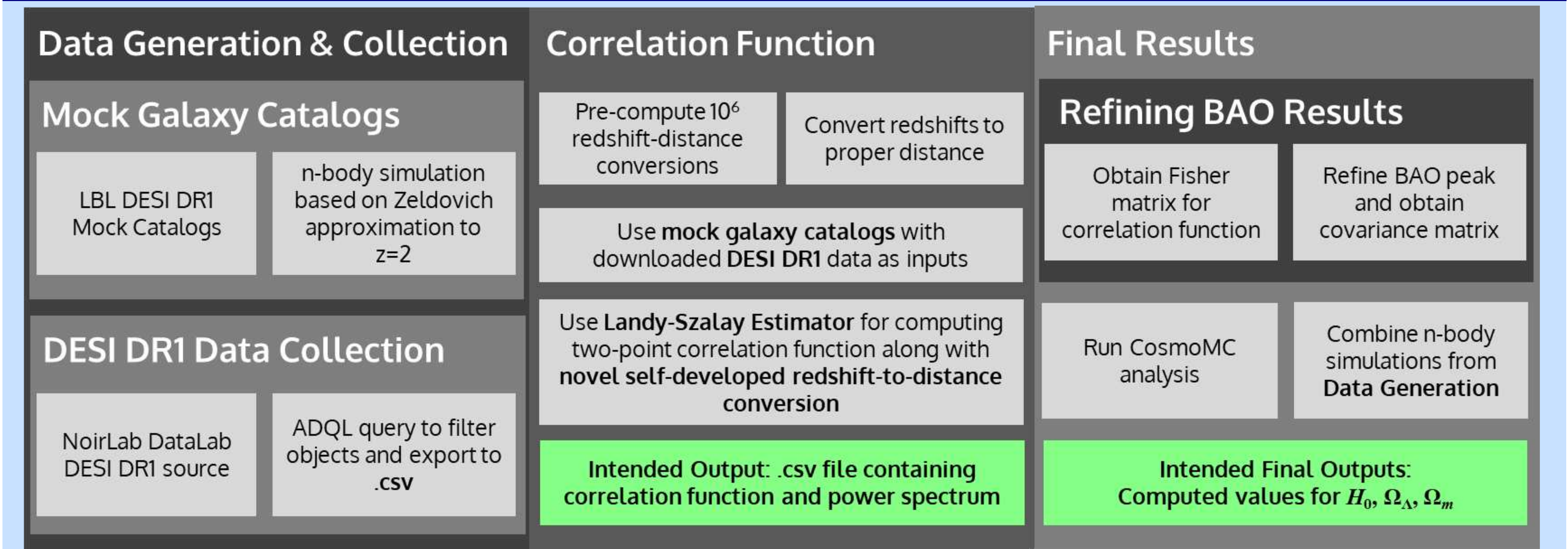
However, there is a **lack of studies** concerning the the detection of the BAO signal at **high redshift**, and the use of the DESI DR1 as a dataset for BAO analysis.

Furthermore, there is a large **2-sigma tension** between computed values of H_0 from **CMB-based** and **observational results**, where values conflict between **68** and **72 $\text{km s}^{-1} \text{Mpc}^{-1}$** .

Therefore, a **thorough analysis** of DESI DR1 at high redshift is necessary to **detect the BAO signal** and **compute the Hubble constant** at high redshift in an effort to help **resolve** the Hubble tension.

Uncovering the Secrets of the Universe: Hubble Constant Found from High-Redshift Baryon Acoustic Oscillations

Methods



Correlation Function

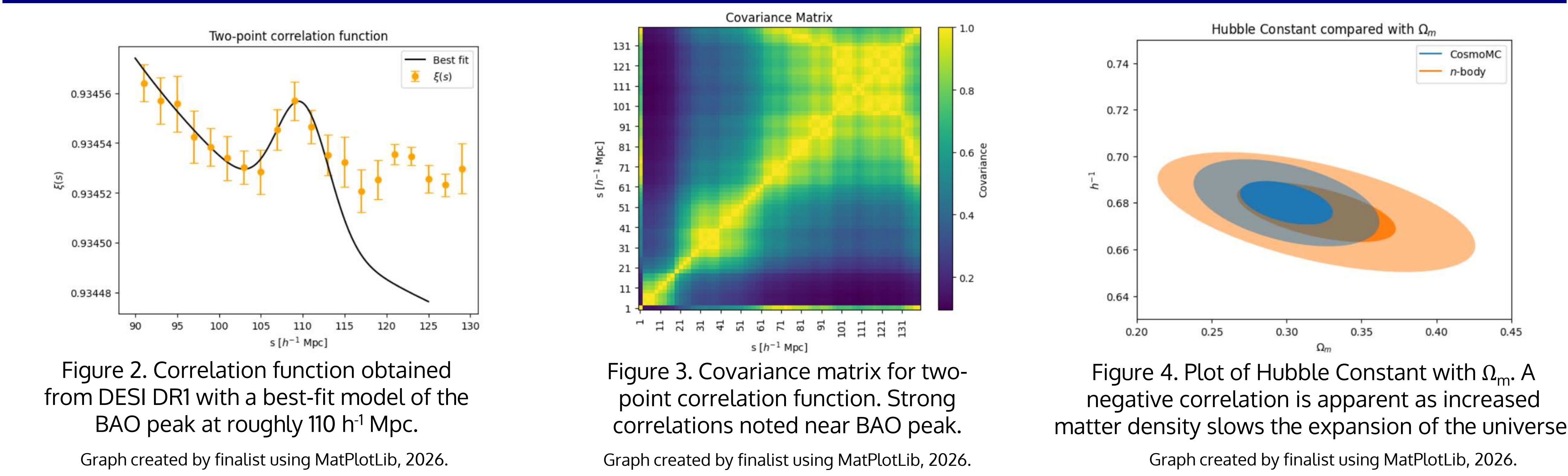
Landy-Szalay Estimator
(Landy and Szalay, 1993)

$$\xi(s) = \frac{DD(s) - 2DR(s) + RR(s)}{RR(s)}$$

Redshift to Distance Computation

$$\chi(z) = \int_1^{1+z} (\Omega_m x^3 + (1 - \Omega_m - \Omega_\Lambda) x^2 + \Omega_\Lambda)^{-\frac{1}{2}} dx$$
$$P^2 = \left(\frac{1}{1+z_1} q_1 \right)^2 \cdot 2 \left(\left(\Sigma \left(\frac{\chi(z_2) + \chi(z_1)}{2}, 1 \right) \right)^2 \sin^2 \left(\frac{\alpha}{2} \right) + \left(\Sigma \left(\frac{\chi(z_2) - \chi(z_1)}{2}, 1 \right) \right)^2 \cos^2 \left(\frac{\alpha}{2} \right) \right)^{\frac{1}{2}} \quad (\text{Liske, 2000})$$
$$z'_2 = \frac{2P^2}{(q_1 - 2P^2)^2} \left(1 + q_1 - 2P^2 + \sqrt{\frac{q_1^2}{P^2} + 1 - 2q_1} \right) \quad (\text{Liske, 2000})$$
$$\Sigma(\chi(z), k) = \begin{cases} \frac{\chi(z)}{\sin(k\chi(z))} & k = 1 \\ \frac{\sqrt{k}}{\sinh(k\chi(z))} & k < 1 \\ \frac{\sqrt{k}}{\sqrt{k}} & k > 1 \end{cases}$$
$$q_1(z) = -1 + \frac{3(\Omega_m)(1+z)^3}{2((\Omega_m)(1+z)^3 + (1 - \Omega_m))}$$

Results



Computed Cosmological Parameters

$H_0 = 67.8 \pm 1.2 \text{ km s}^{-1} \text{Mpc}^{-1}$

$\Omega_m = 0.31 \pm 0.084$

$\Omega_\Lambda = 0.69 \pm 0.084$

Figure 5. Plot of Ω_Λ with Ω_m . There is a negative correlation, which is expected as $\Omega_\Lambda + \Omega_m = 1$ in a Λ CDM universe.

Graph created by finalist using Matplotlib, 2026.

Discussion

As seen in Figures 2 and 3, it is strongly implied that the peak detected at **110 $h^{-1} \text{Mpc}$ is a true BAO peak**. This figure is relatively close to the established value of about **105 $h^{-1} \text{Mpc}$** , implying that the BAO peak detected is an accurate measurement.

Detection Importance

- **Confirmed BAO detection** with DESI DR1
- Use of BAOs as a **standard ruler at high redshift**
- Further results **aligning with CMB-based data**
- **More data** to help **resolve Hubble tension**

The best-fit cosmological parameters of $H_0 = 67.8 \pm 1.2 \text{ km s}^{-1} \text{Mpc}^{-1}$, $\Omega_\Lambda = 0.69 \pm 0.084$, and $\Omega_m = 0.31 \pm 0.084$ supports the CMB-based data and aligns with the Λ CDM model, visualized in Figures 4 and 5. This result may suggest that the Hubble constant at higher redshifts based on observational data may also align with such measurements.

Future Work

- Computation of **other cosmological parameters**, such as radius of the sound horizon at photon decoupling
- Possible inclusion of **machine-learning** for quickly evolving mock CMB power spectra to a final state

References

Eisenstein, D. J., Zehavi, I., Hogg, D. W., Scoccimarro, R., Blanton, M. R., Nichol, R., Scranton, R. C., Seo, H., Tegmark, M., Zheng, Z., Anderson, S. F., Annis, J., Bahcall, N., Brinkmann, J., Burles, S., Castander, F. J., Connolly, A., Csabai, I., Doi, M., Hennessy, G., Ivezić, Z., Kent, S., Knapp, G. R., Lin, H., Loh, Y., Lupton, R. H., Margon, B., McKay, T. A., Meiksin, A., Munn, J. A., Pope, A., Richmond, M. W., Schlegel, D., Schneider, D. P., Shimasaku, K., Stoughton, C., Strauss, M. A., SubbaRao, M., Szalay, A. S., Szapudi, I., Tucker, D. L., Yanny, B., & York, D. G. (2005) Detection of the Baryon Acoustic Peak in the Large-Scale Correlation Function of SDSS Luminous Red Galaxies. The Astrophysical Journal, 633(2), 560-574 <https://doi.org/10.1086/466512>

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